BharatWiki AI: Intelligent Multilingual Knowledge Discovery Platform

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DECLARATION

The Mini Project Report entitled "BharatWiki AI: Intelligent Multilingual Knowledge Discovery Platform" is a record of bonafide work of D. Shreenitha – 2520030504, A. Srihitha – 2520030566, and G. Neha – 252030068, submitted in partial fulfillment for the award of B. Tech in Computer Science and Engineering to K L (Deemed to be) University. The results embodied in this report have not been copied from any other department, university, or institute.

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CERTIFICATE

This is to certify that the mini project report entitled "BharatWiki AI: Intelligent Multilingual Knowledge Discovery Platform" is a bonafide work done and submitted by D. Shreenitha – 2520030504, A. Srihitha – 2520030566, and G. Neha – 252030068 in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering, K L (Deemed to be) University, during the academic year 2026–2027.

Signature of the Supervisor

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1.ABSTRACT

BharatWiki AI is an AI-powered multilingual search and knowledge discovery platform designed to provide fast and intelligent access to Wikipedia content in Indian languages, initially Hindi, Telugu, Tamil, and Kannada. Wikipedia Dumps are processed using text cleaning, Unicode normalization, tokenization, stop-word removal, and indexing. An inverted index and Apache Lucene support efficient retrieval, while TF-IDF and BM25 are used to rank documents by relevance. The platform is designed to support single-keyword, multiple-keyword, and multilingual searches. NLP and AI components are intended to provide semantic search, AI-generated summaries, related-article recommendations, search suggestions, and keyword highlighting. The proposed technology stack includes React for the frontend, Java Spring Boot for the backend, PostgreSQL for data storage, Apache Lucene for searching and indexing, and NLP/AI components for intelligent features. The expected interface presents ranked results with summaries and related information. An analytics dashboard is intended to display search trends, popular topics, language distribution, and response time. The project demonstrates how data structures, information retrieval, multilingual computing, NLP, and AI can be combined to make knowledge discovery faster and more useful for Indian-language users. The platform may later be extended to government documents, research papers, books, news articles, PDFs, and educational content.

Team Name: BharatWiki AI

Team Logo (if any): Not provided

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2. INTRODUCTION

BharatWiki AI addresses the difficulty of finding relevant information across large collections of multilingual Wikipedia content. Conventional keyword searching may not provide well-ranked results, summaries, or related information, particularly when users search across Indian languages. The project proposes a knowledge discovery platform for Hindi, Telugu, Tamil, and Kannada Wikipedia content. Wikipedia Dumps are preprocessed through text cleaning, Unicode normalization, tokenization, and stop-word removal before indexing. Apache Lucene and an inverted index support efficient search, while TF-IDF and BM25 help rank results. NLP and AI components are planned to support semantic search, summaries, recommendations, suggestions, and keyword highlighting. This report presents the project background, related research, problem statement, requirements, implementation approach, results to be evaluated, and future scope.

The project brings together data structures, information retrieval, multilingual processing, and AI to make knowledge discovery more accessible. Its initial focus is Indian-language Wikipedia, with the possibility of extending the same approach to other information sources.

3. LITERATURE SURVEY

1. Cross-Lingual Information Retrieval System for Indian Languages (Jagarlamudi and Kumaran, 2008): Studies retrieval across Indian languages and provides relevant background for the multilingual search goal of BharatWiki AI.

2. Exploiting Wikipedia for Cross-Lingual and Multilingual Information Retrieval (2012): Explores Wikipedia as a multilingual knowledge resource and relates to the project's use of multiple Wikipedia language editions.

3. Exploiting Wikipedia API for Hindi-English Cross-language Information Retrieval (2016): Investigates Wikipedia-based cross-language retrieval and relevance scoring approaches, including BM25, relevant to the project's ranking approach.

4. Towards Building a Multilingual Semantic Network: Identifying Interlingual Links in Wikipedia (Dandala, Mihalcea, and Bunescu, 2012): Studies semantic relationships through interlingual links and informs multilingual knowledge discovery.

5. WikiTranslate: Query Translation for Cross-lingual Information Retrieval using only Wikipedia: Uses Wikipedia concepts and translations to support cross-language queries.

6. Analysing Entity Context in Multilingual Wikipedia to Support Entity-Centric Retrieval Applications (2016): Examines entity information across language editions and relates to related-article discovery.

7. Considerations for Multilingual Wikipedia Research (Johnson and Lescak, 2022): Discusses differences between Wikipedia language editions and dataset challenges, highlighting the importance of careful preprocessing.

4. Problem Statement and Objectives

Problem Statement: A clear and specific articulation of the problem, outlining its importance. Searching large amounts of Indian-language Wikipedia content efficiently and retrieving relevant, understandable results is challenging.

Problem Description: A concise explanation of the issue or challenge that needs to be solved. Traditional keyword search may not rank results intelligently or provide summaries, related articles, and cross-language discovery. BharatWiki AI aims to improve access through indexing, relevance ranking, and NLP/AI features.

Objectives:

- Provide search across Hindi, Telugu, Tamil, and Kannada Wikipedia content.
- Improve retrieval speed using an inverted index and Apache Lucene.
- Rank documents using TF-IDF and BM25.
- Support multilingual and semantic search with NLP/AI components.
- Present summaries, related-article recommendations, search suggestions, and keyword highlighting.
- Provide analytics on search trends, popular topics, language distribution, and response time.

5. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements:

• Computer/laptop with a modern multi-core processor
• 8 GB RAM recommended for processing Wikipedia data and running the search stack
• Sufficient storage for Wikipedia Dumps, indexes, database files, and dependencies
• Internet connection for obtaining datasets and development resources

Software Requirements:

• Java / JDK
• Node.js and npm
• React.js
• Java Spring Boot
• PostgreSQL
• Apache Lucene
• NLP/AI components
• Wikipedia Dumps
• VS Code / IntelliJ IDEA

6. IMPLEMENTATION

Implementation modules:

Module 1 — Dataset preprocessing: Load Wikipedia Dumps, clean text, normalize Unicode, tokenize content, remove stop words, and prepare documents for indexing**.**

Module 2 — Inverted index and search: Use Apache Lucene to build and query an index so matching documents can be retrieved efficiently.

Module 3 — Relevance ranking: Use TF-IDF/BM25 relevance scoring to order retrieved documents according to their match with the user's query.

Module 4 — Intelligent results and interface: The React interface presents ranked results; NLP/AI components are intended to support semantic search, summaries, related-article recommendations, suggestions, keyword highlighting, and analytics.

7. RESULTS & DISCUSSION

Add screenshots from the actual application run here. Suggested screenshots are: (1) multilingual search form and language selection; (2) ranked results with highlighted keywords; (3) summary and related-article panel; and (4) analytics dashboard. Each screenshot should have a two-line explanation. Actual screenshots were not included in the provided files.

8. CONCLUSION & FUTURE SCOPE

BharatWiki AI proposes a unified platform for searching multilingual Indian-language Wikipedia content. Its design combines preprocessing, inverted indexing, Apache Lucene, TF-IDF/BM25 ranking, and NLP/AI-assisted features to improve discovery of relevant information. The project objectives include faster retrieval, ranked results, multilingual access, summaries, recommendations, and analytics. The effectiveness of these features should be assessed using the completed application and measured search results.

Future Scope:

• Extend the supported languages and improve cross-language query handling.

• Add more Wikipedia language editions and larger datasets.

• Expand beyond Wikipedia to government documents, research papers, books, news articles, PDFs, and educational content.

• Evaluate relevance, summary quality, and response time using test queries.

• Add user feedback, accessibility features, and richer analytics.

9. REFERENCES

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